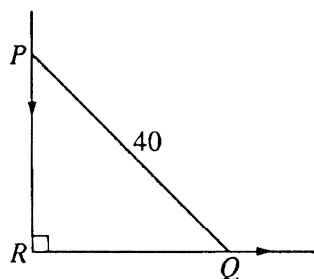


- BC 37. A person 2 meters tall walks directly away from a streetlight that is 8 meters above the ground. If
1988 the person is walking at a constant rate and the person's shadow is lengthening at the rate of $\frac{4}{9}$
NC meter per second, at what rate, in meters per second, is the person walking?

(A) $\frac{4}{27}$ (B) $\frac{4}{9}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$ (E) $\frac{16}{9}$

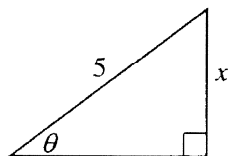
- BC 4. A particle moves along the curve $xy = 10$. If $x = 2$ and $\frac{dy}{dt} = 3$, what is the value of $\frac{dx}{dt}$?
1993

SC (A) $-\frac{5}{2}$ (B) $-\frac{6}{5}$ (C) 0 (D) $\frac{4}{5}$ (E) $\frac{6}{5}$



- BC 34. In the figure above, PQ represents a 40-foot ladder with end P against a vertical wall and end Q on
1993 level ground. If the ladder is slipping down the wall, what is the distance RQ at the instant when Q
SC is moving along the ground $\frac{3}{4}$ as fast as P is moving down the wall?

(A) $\frac{6}{5}\sqrt{10}$ (B) $\frac{8}{5}\sqrt{10}$ (C) $\frac{80}{\sqrt{7}}$ (D) 24 (E) 32



- BC 23. In the triangle shown above, if θ increases at a constant rate of 3 radians per minute, at what rate is
1997 x increasing in units per minute when x equals 3 units?
NC

(A) 3 (B) $\frac{15}{4}$ (C) 4 (D) 9 (E) 12